

Multithreading

Study Guide

Mr. Shivkumar Lilhare
Assistant Professor (CSE), PIT
Parul University

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9.2.1 Types of Threads

✓ 1. Single-threaded vs. Multithreaded Programs

- **Single-threaded:**
 - Executes one task at a time.
 - Simpler and easier to debug but inefficient for multitasking.
- **Multithreaded:**
 - Multiple threads execute concurrently.
 - Efficient CPU utilization.
 - Example: Web server handling multiple requests simultaneously.

✓ 2. Main Thread and Child Threads

- **Main Thread:**
 - The first thread that starts when a Java program begins execution.
 - Created automatically by the JVM.

```
public class MainThreadExample {  
    public static void main(String[] args) {  
        System.out.println("Main thread: " + Thread.currentThread().getName());  
    }  
}
```

- **Child Threads:**
 - Created explicitly using Thread class or Runnable interface.
 - Run in parallel with the main thread.

9.2.2 Thread Communication Methods

◆ sleep()

- Pauses the current thread for a specified time.
- Syntax: Thread.sleep(milliseconds)
- Throws InterruptedException.
- Syntax:

```
Thread.sleep(1000); // Sleeps for 1 second
```

◆ **yield()**

- Causes the currently executing thread to pause and allow other threads of equal priority to execute.
- Doesn't guarantee another thread will be scheduled.
- Syntax:

```
Thread.yield();
```

◆ **join()**

- Waits for a thread to finish execution before the next code continues.

```
t1.join(); // Waits for t1 to complete
```

◆ **wait() and notify()/notifyAll()**

- Used for **inter-thread communication**.
- Must be called from **synchronized blocks or methods**.
- wait(): Thread waits and releases the lock.
- notify(): Wakes up a single thread waiting on that object's monitor.
- notifyAll(): Wakes up all waiting threads.

```
synchronized(obj) {  
    obj.wait();  
    obj.notify();  
}
```

9.2.3 Thread Safety

✓ **What is Thread Safety?**

- **Thread safety** ensures that shared data is accessed and modified by multiple threads in a predictable and controlled manner.
- Prevents **race conditions**, **data inconsistency**, and **unexpected behavior** in multithreaded programs.

✓ **Synchronization in Java**

- Synchronization allows only one thread at a time to access a **critical section** (shared resource).
- Helps ensure **mutual exclusion** and **data consistency**.

✓ Synchronized Methods and Blocks

◆ Synchronized Method

- Locks the entire method for the calling object.

```
public synchronized void increment() {  
    count++;  
}
```

◆ Synchronized Block

- Synchronizes a specific portion of code using an object as a lock.

```
synchronized(this) {  
    count++;  
}
```

Use synchronized blocks when you want to lock only a critical section rather than the whole method.

✓ volatile Keyword

- Tells the JVM that a variable may be modified by multiple threads.
- Ensures **visibility**, not atomicity.
- Prevents thread from caching the variable locally.

```
private volatile boolean running = true;
```

Note: Good for flags and state variables that are read/written by multiple threads.

➤ Summary of Key Concepts:

- Java can run both single-threaded and multithreaded applications.
- Main thread is created by JVM; child threads are user-created.
- sleep(), yield(), join() help in controlling thread execution timing.
- wait() and notify() enable threads to communicate and coordinate execution.
- All thread communication methods are essential for smooth concurrency.
- Thread safety is essential to prevent data corruption in multithreaded programs.
- Synchronization ensures mutual exclusion using synchronized methods or blocks.
- volatile provides visibility but not atomicity.

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